

ULS combinations of actions on a reinforced concrete retaining wall

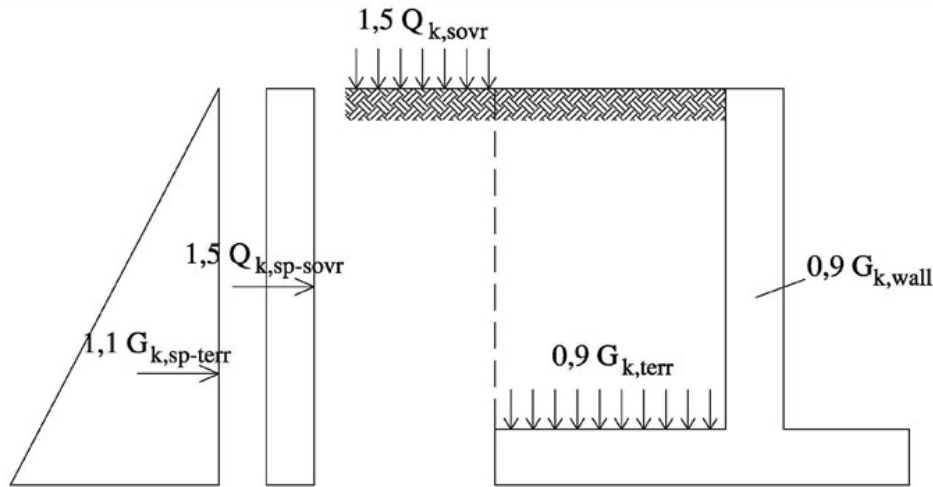


Fig. 2.9. Actions for EQU ULS verification of a retaining wall in reinforced concrete

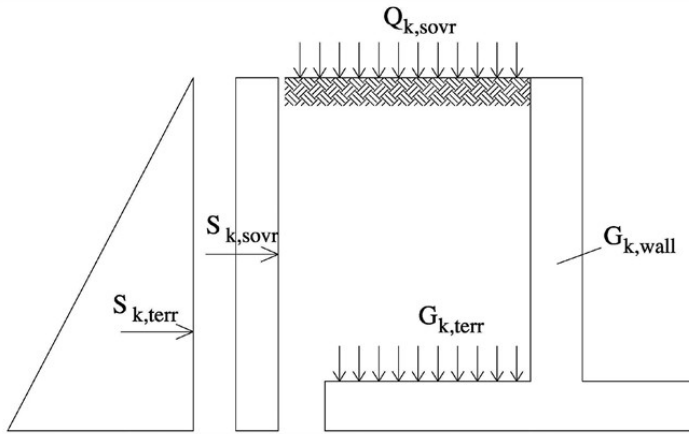


Fig. 2.8. Actions on a retaining wall in reinforced concrete

Only that part of the embankment beyond the foundation footing is considered for the verification of global stability to heave and sliding

$$1.1 \cdot S_{k,terr} + 0.9 \cdot (G_{k,wall} + G_{k,terr}) + 1.5 \cdot S_{k,sovr}$$

Approach 1

Design values from Set C and from Set B are applied in separate calculations to the geotechnical actions and to all other actions from the structure or on the structure.

Set C

$$1 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.3 \cdot S_{k,sovr}$$

Set B

$$1.35 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

$$1.35 \cdot S_{k,terr} + 1.35 \cdot G_{k,wall} + 1.35 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

$$1.35 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1.35 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

$$1.35 \cdot S_{k,terr} + 1.35 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

Note: For all the above-listed combinations, two possibilities must be considered: either that the surcharge concerns only the part of embankment beyond the foundation footing (Fig. 2.10a), or that it acts on the whole surface of the embankment (Fig. 2.10b).

For brevity, only cases in relation with case b), i.e. with surcharge acting on the whole surface of embankment, are given below.

The following figures show loads in relation to the combinations obtained with Set B partial safety factors.

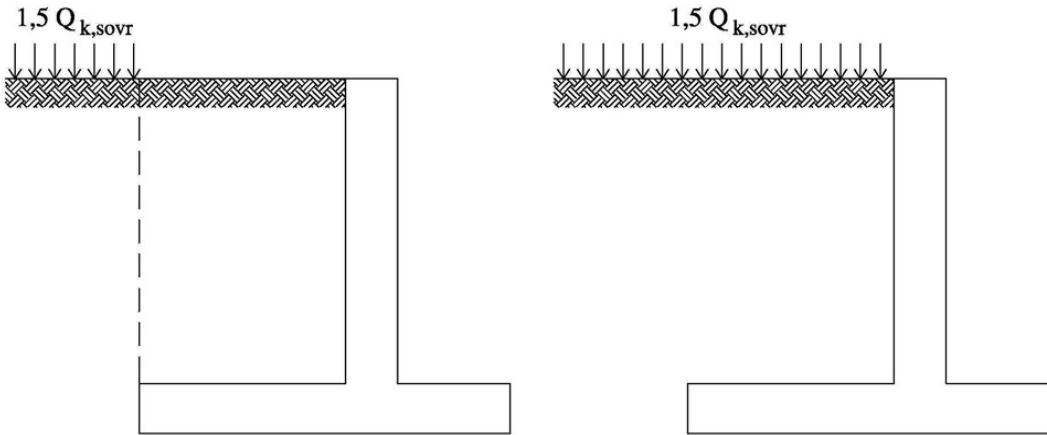


Fig. 2.10. Possible load cases of surcharge on the embankment.

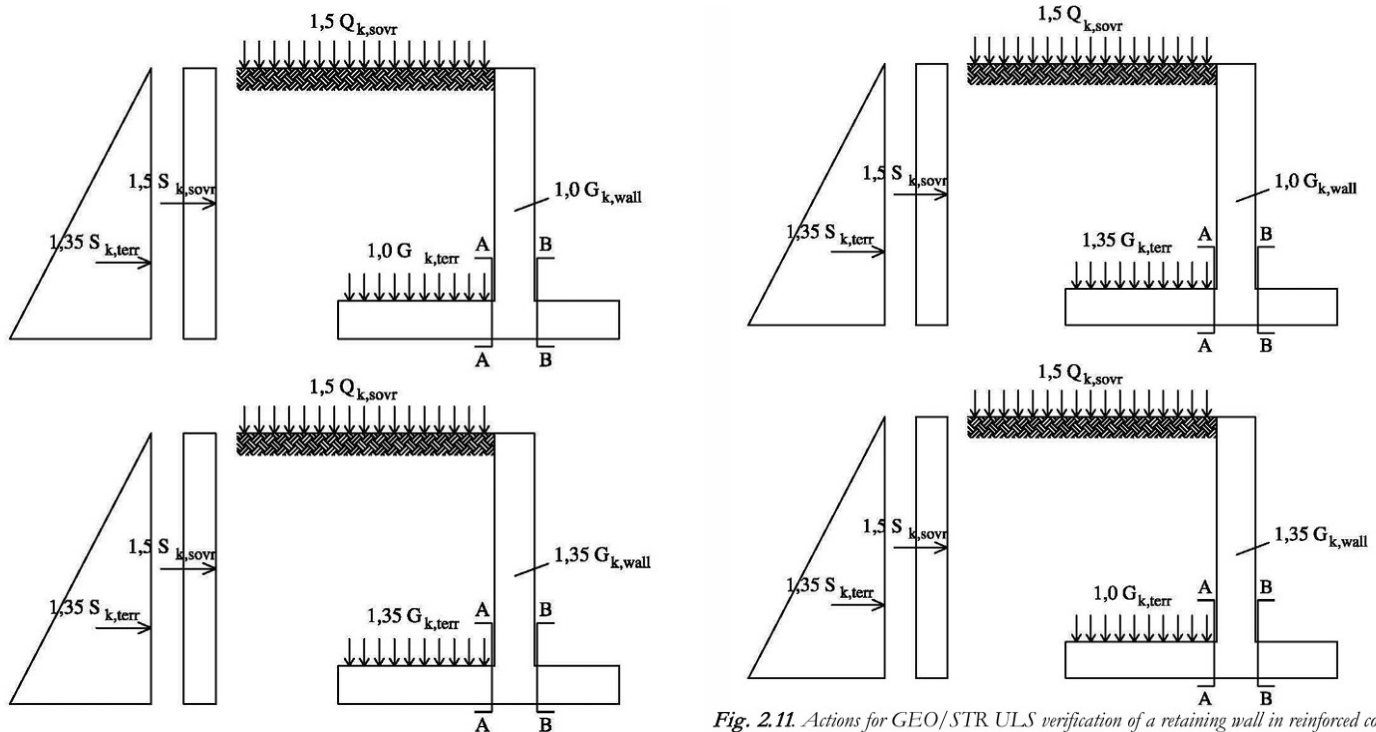


Fig. 2.11. Actions for GEO/STR ULS verification of a retaining wall in reinforced concrete.

Approach 2

Set B is used

$$1.35 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

$$1.35 \cdot S_{k,terr} + 1.35 \cdot G_{k,wall} + 1.35 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

$$1.35 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1.35 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

$$1.35 \cdot S_{k,terr} + 1.35 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.5 \cdot S_{k,sovr} + 1.5 \cdot Q_{k,sovr}$$

Approach 3

Factors from Set C for geotechnical actions and from Set B for other actions are used in one calculation.

$$1 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.3 \cdot S_{k,sovr} + 1.3 \cdot Q_{k,sovr}$$

$$1 \cdot S_{k,terr} + 1.35 \cdot G_{k,wall} + 1.35 \cdot G_{k,terr} + 1.3 \cdot S_{k,sovr} + 1.3 \cdot Q_{k,sovr}$$

$$1 \cdot S_{k,terr} + 1 \cdot G_{k,wall} + 1.35 \cdot G_{k,terr} + 1.3 \cdot S_{k,sovr} + 1.3 \cdot Q_{k,sovr}$$

$$1 \cdot S_{k,terr} + 1.35 \cdot G_{k,wall} + 1 \cdot G_{k,terr} + 1.3 \cdot S_{k,sovr} + 1.3 \cdot Q_{k,sovr}$$

EXAMPLE Retaining structures

The assumption is initially made that the surcharge acts only on the part of embankment beyond the foundation footing.

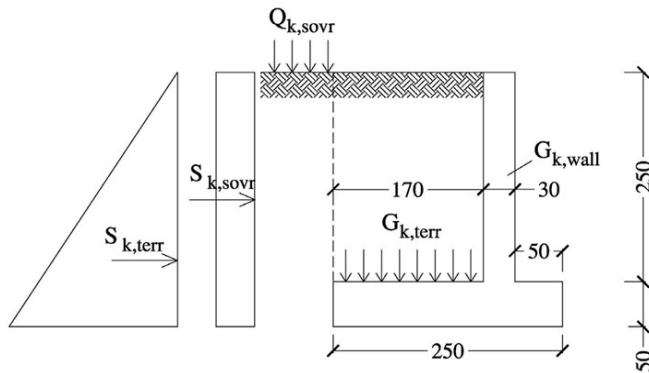


Fig. 2.12. Wall dimensions and actions on the wall (surcharge outside the foundation footing).

$$\gamma := 18 \frac{\text{kN}}{\text{m}^3} \quad \text{weight density}$$

$$\varphi := 30^\circ \quad \text{angle of shearing resistance}$$

$$K_a := 0.33 \quad \text{factor of horiz. active earth pressure:}$$

$$\delta := 0^\circ \quad \text{wall-ground interface friction angle}$$

$$\mu_{\text{groundflooring}} := 0.57 \quad \text{ground-flooring friction factor}$$

$$\rho := 25 \frac{\text{kN}}{\text{m}^3} \quad t_w := 0.3 \text{ m} \quad b := 2.5 \text{ m} \quad T_f := 0.5 \text{ m} \quad h_w := 3 \text{ m} \quad b_{f1} := 1.7 \text{ m} \quad b_{f2} := 0.5 \text{ m}$$

$$P_{k,wall} := t_w \cdot b \cdot \rho = 18.75 \frac{\text{kN}}{\text{m}} \quad \text{self-weight of wall}$$

$$P_{k,foot} := T_f \cdot b \cdot \rho = 31.25 \frac{\text{kN}}{\text{m}} \quad \text{self-weight of footing}$$

$$G_{k,wall} := P_{k,wall} + P_{k,foot} = 50 \frac{\text{kN}}{\text{m}}$$

$$G_{k,ground} := b_{f1} \cdot b \cdot \gamma = 76.5 \frac{\text{kN}}{\text{m}} \quad \text{self weight of ground above footing}$$

$$Q_{k,surch} := 10 \frac{\text{kN}}{\text{m}^2} \quad \text{surcharge on embankment}$$

$$S_{k,ground} := 26.73 \frac{\text{kN}}{\text{m}} \quad \text{ground horizontal force}$$

$$S_{k.surch} := 9.9 \frac{\text{kN}}{\text{m}}$$

surcharge horizontal force

Verification to failure by sliding

Slide force

Ground horizontal force ($\gamma_G = 1.1$): $\gamma_g := 1.1$

$$S_{ground} := \gamma_g \cdot S_{k.ground} = 29.403 \frac{\text{kN}}{\text{m}}$$

Surcharge horizontal ($\gamma_Q = 1.5$) $\gamma_q := 1.5$

$$S_{sur} := \gamma_q \cdot S_{k.surch} = 14.85 \frac{\text{kN}}{\text{m}}$$

Sliding force

$$F_{slide} := S_{sur} + S_{ground} = 44.253 \frac{\text{kN}}{\text{m}}$$

Resistant force (in the assumption of ground-flooring friction factor = 0.57)

$$\gamma_g := 0.9$$

wall self-weight ($\gamma_G = 0.9$)

$$F_{stab.wall} := \gamma_g \cdot (P_{k.wall} \cdot \mu_{groundflooring}) = 9.6188 \frac{\text{kN}}{\text{m}}$$

footing self-weight ($\gamma_G = 0.9$)

$$F_{stab.foot} := \gamma_g \cdot (P_{k.foot} \cdot \mu_{groundflooring}) = 16.0312 \frac{\text{kN}}{\text{m}}$$

ground self-weight ($\gamma_G = 0.9$)

$$F_{stab.ground} := \gamma_g \cdot (G_{k.ground} \cdot \mu_{groundflooring}) = 39.2445 \frac{\text{kN}}{\text{m}}$$

resistant force

$$F_{stab} := F_{stab.wall} + F_{stab.foot} + F_{stab.ground} = 64.8945 \frac{\text{kN}}{\text{m}}$$

The safety factor for sliding is:

$$FS := \frac{F_{stab}}{F_{slide}} = 1.4664$$

Verification to Overturning

overturning moment

Ground horizontal force ($\gamma_G = 1.1$): $\gamma_g := 1.1$

$$M_{s.ground} := \gamma_g \cdot \left(\frac{S_{k.ground} \cdot h_w}{3} \right) = 29.403 \text{ m} \frac{\text{kN}}{\text{m}}$$

Surcharge horizontal ($\gamma_Q = 1.5$) $\gamma_q := 1.5$

$$M_{s.surch} := \gamma_q \cdot \left(\frac{S_{k.surch} \cdot \frac{h_w}{2}}{1} \right) = 22.275 \text{ m} \frac{\text{kN}}{\text{m}}$$

Sliding force

$$M_{rib} := M_{s.ground} + M_{s.surch} = 51.678 \text{ m} \frac{\text{kN}}{\text{m}}$$

stabilizing moment (in the assumption of ground-flooring friction factor = 0.57)

$$\gamma_g := 0.9$$

moment wall self-weight ($\gamma_g = 0.9$):

$$M_{stab.wall} := \gamma_g \cdot \left(P_{k.wall} \cdot \left(b_{f2} + \frac{t_w}{2} \right) \right) = 10.9688 \text{ m} \frac{\text{kN}}{\text{m}}$$

moment footing self-weight ($\gamma_g = 0.9$):

$$M_{stab.foot} := \gamma_g \cdot \left(P_{k.foot} \cdot \frac{b}{2} \right) = 35.1562 \text{ m} \frac{\text{kN}}{\text{m}}$$

moment wall self-weight ($\gamma_g = 0.9$):

$$M_{stab.ground} := \gamma_g \cdot \left(G_{k.ground} \cdot \left(b_{f2} + t_w + \frac{b_{f1}}{2} \right) \right) = 113.6025 \text{ m} \frac{\text{kN}}{\text{m}}$$

stabilizing moment

$$M_{stab} := M_{stab.wall} + M_{stab.foot} + M_{stab.ground} = 159.7275 \text{ m} \frac{\text{kN}}{\text{m}}$$

safety factor to global stability:

$$FS := \frac{M_{stab}}{M_{rib}} = 3.0908$$

Contact pressure on ground

Approach 2, i.e. Set B if partial factors, is used.

By taking 1.0 and 1.35 as the partial factors for the self-weight of the wall and of the ground above the foundation footing respectively, we obtain four different combinations as seen above:

$$1.35 \cdot S_{k.terr} + 1 \cdot G_{k.wall} + 1 \cdot G_{k.terr} + 1.5 \cdot S_{k.sovr} + 1.5 \cdot Q_{k.sovr}$$

$$1.35 \cdot S_{k.terr} + 1.35 \cdot G_{k.wall} + 1.35 \cdot G_{k.terr} + 1.5 \cdot S_{k.sovr} + 1.5 \cdot Q_{k.sovr}$$

$$1.35 \cdot S_{k.terr} + 1 \cdot G_{k.wall} + 1.35 \cdot G_{k.terr} + 1.5 \cdot S_{k.sovr} + 1.5 \cdot Q_{k.sovr}$$

$$1.35 \cdot S_{k.terr} + 1.35 \cdot G_{k.wall} + 1 \cdot G_{k.terr} + 1.5 \cdot S_{k.sovr} + 1.5 \cdot Q_{k.sovr}$$

the contact pressure on ground is calculated, for the first of the fourth above-mentioned combinations, as follows:

moment vs. centre of mass of the footing

moment from ground lateral force ($\gamma_g = 1.35$):

$$\gamma_g := 1.35$$

$$M_{s.terr} := \gamma_g \cdot \left(\frac{S_{k.ground} \cdot h_w}{3} \right) = 36.0855 \text{ kN} \frac{\text{m}}{\text{m}}$$

$$\gamma_q := 1.5$$

moment from surcharge lateral force ($\gamma_q = 1.5$):

$$M_{s.sovr} := \gamma_q \cdot \left(\frac{S_{k.surch} \cdot \frac{h_w}{2}}{1} \right) = 22.275 \text{ kN} \frac{\text{m}}{\text{m}}$$

$$\gamma_g := 1$$

moment from wall self-weight ($\gamma_g = 1.0$):

$$M_{wall} := \gamma_g \cdot \left(\frac{P_{k.wall} \cdot 0.6 \text{ m}}{1} \right) = 11.25 \text{ kN} \frac{\text{m}}{\text{m}}$$

moment from footing self-weight ($\gamma_g = 1.0$):

$$\gamma_g := 1$$

$$M_{foot} := \gamma_g \cdot (0) = 0 \text{ kN} \frac{\text{m}}{\text{m}}$$

moment from ground self-weight ($\gamma_G = 1.0$):

$$\gamma_g := 1$$

$$M_{ground} := (-1) \cdot \gamma_g \cdot \left(\frac{G_{k,ground} \cdot 0.4 \text{ m}}{1} \right) = -30.6 \text{ kN} \frac{\text{m}}{\text{m}}$$

$$M_{tot} := M_{s,terr} + M_{s,sovr} + M_{wall} + M_{foot} + M_{ground} = 39.0105 \text{ kN} \frac{\text{m}}{\text{m}}$$

Vertical load

Wall self-weight ($\gamma_G = 1.0$):

$$\gamma_g := 1$$

$$P_{wall} := \gamma_g \cdot P_{k,wall} = 18.75 \text{ kN} \frac{\text{m}}{\text{m}}$$

Footing self-weight ($\gamma_G = 1.0$):

$$\gamma_g := 1$$

$$P_{foot} := \gamma_g \cdot P_{k,foot} = 31.25 \text{ kN} \frac{\text{m}}{\text{m}}$$

Ground self-weight ($\gamma_G = 1.0$):

$$\gamma_g := 1$$

$$P_{ground} := \gamma_g \cdot G_{k,ground} = 76.5 \text{ kN} \frac{\text{m}}{\text{m}}$$

$$P_{tot} := P_{wall} + P_{foot} + P_{ground} = 126.5 \text{ kN} \frac{\text{m}}{\text{m}}$$

Eccentricity

$$e := \frac{M_{tot}}{P_{tot}} = 0.3084 \text{ m}$$

$$B := 2.5 \text{ m}$$

$$\frac{B}{6} = 0.4167 \text{ m}$$

$$\text{if } e \leq \frac{B}{6}$$

result := "good"

else

result := "bad"

result = "good"

The results given at Table 2.2 are obtained by repeating the calculation for the three remaining combinations of partial factors.

The maximal pressure on ground is achieved with the second combination, i.e. for the one in which the wall self-weight and the self-weight of the ground above the footing are both multiplied by 1.35.

For the verification of the contact pressure, the possibility that the surcharge acts on the whole embankment surface must be also considered. (Fig. 2.13); the values given at Table 2.3 are obtained by repeating the calculation for this situation.

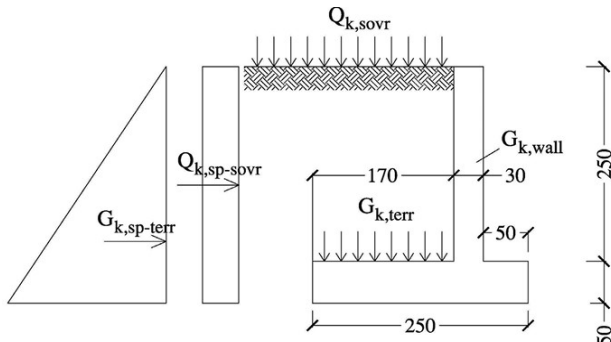


Fig. 2.13. Dimensions of the retaining wall of the numeric example with surcharge on the whole embankment.

Table 2.2. Max pressure for four different combinations of partial factors of permanent loads (surcharge outside the foundation footing).

Combination	first	second	third	fourth
$M_{s,ground}$ (kNm/m)	36.08 ($\gamma_Q=1.35$)	36.08 ($\gamma_Q=1.35$)	36.08 ($\gamma_Q=1.35$)	36.08 ($\gamma_Q=1.35$)
$M_{s,surch}$ (kNm/m)	22.28 ($\gamma_Q=1.5$)	22.28 ($\gamma_Q=1.5$)	22.28 ($\gamma_Q=1.5$)	22.28 ($\gamma_Q=1.5$)
M_{wall} (kNm/m)	11.25 ($\gamma_G=1.0$)	15.19 ($\gamma_G=1.35$)	11.25 ($\gamma_G=1.0$)	15.19 ($\gamma_G=1.35$)
M_{ground} (kNm/m)	-30.60 ($\gamma_G=1.0$)	-41.31 ($\gamma_G=1.35$)	-41.31 ($\gamma_G=1.35$)	-30.60 ($\gamma_G=1.0$)
M_{tot} (kNm/m)	39.01	32.24	28.30	42.95
P_{wall} (kN/m)	18.75 ($\gamma_G=1.0$)	25.31 ($\gamma_G=1.35$)	18.75 ($\gamma_G=1.0$)	25.31 ($\gamma_G=1.35$)
P_{foot} (kN/m)	31.25 ($\gamma_G=1.0$)	42.19 ($\gamma_G=1.35$)	31.25 ($\gamma_G=1.0$)	42.19 ($\gamma_G=1.35$)
P_{ground} (kN/m)	76.50	103.28	103.28	76.50
(kN/m)	($\gamma_G=1.0$)	($\gamma_G=1.35$)	($\gamma_G=1.35$)	($\gamma_G=1.0$)
P_{tot} (kN/m)	126.50	170.78	153.28	144
eccentricity (m)	0.31	0.19	0.18	0.30
pressure on ground (kN/m ²)	88.05	99.26	88.48	98.83

Table 2.3. Max pressure on ground for four different combinations of partial factors of permanent loads (surcharge on the whole foundation footing).

Combination	first	second	third	fourth
$M_{s,ground}$ (kNm/m)	36.08 ($\gamma_Q=1.35$)	36.08 ($\gamma_Q=1.35$)	36.08 ($\gamma_Q=1.35$)	36.08 ($\gamma_Q=1.35$)
$M_{s,surch}$ (kNm/m)	22.28 ($\gamma_Q=1.5$)	22.28 ($\gamma_Q=1.5$)	22.28 ($\gamma_Q=1.5$)	22.28 ($\gamma_Q=1.5$)
M_{wall} (kNm/m)	11.25 ($\gamma_G=1.0$)	15.19 ($\gamma_G=1.35$)	11.25 ($\gamma_G=1.0$)	15.19 ($\gamma_G=1.35$)
M_{ground} (kNm/m)	-30.60 ($\gamma_G=1.0$)	-41.31 ($\gamma_G=1.35$)	-41.31 ($\gamma_G=1.35$)	-30.60 ($\gamma_G=1.0$)
M_{surch} (kNm/m)	-10.20 ($\gamma_Q=1.5$)	-10.20 ($\gamma_Q=1.5$)	-10.20 ($\gamma_Q=1.5$)	-10.20 ($\gamma_Q=1.5$)
M_{tot} (kNm/m)	28.81	22.04	18.10	32.75
P_{wall} (kN/m)	18.75 ($\gamma_G=1.0$)	25.31 ($\gamma_G=1.35$)	18.75 ($\gamma_G=1.0$)	25.31 ($\gamma_G=1.35$)
P_{foot} (kN/m)	31.25 ($\gamma_G=1.0$)	42.19 ($\gamma_G=1.35$)	31.25 ($\gamma_G=1.0$)	42.19 ($\gamma_G=1.35$)
P_{terr} (kN/m)	76.50 ($\gamma_G=1.0$)	103.28 ($\gamma_G=1.35$)	103.28 ($\gamma_G=1.35$)	76.50 ($\gamma_G=1.0$)
P_{surch} (kN/m)	25.50 ($\gamma_Q=1.5$)	25.50 ($\gamma_Q=1.5$)	25.50 ($\gamma_Q=1.5$)	25.50 ($\gamma_Q=1.5$)
P_{tot} (kN/m)	152.0	196.28	178.78	169.50
eccentricity (m)	0.19	0.11	0.10	0.19
pressure on ground (kN/m ²)	88.46	99.67	88.89	99.24

The two additional lines, not present in Table 2.2 and here highlighted in bold, correspond to the moment and to the vertical load resulting from the surcharge above the footing.

The max pressure on ground is achieved once again for the second combination and its value is here higher than the one calculated in the previous scheme.